

LESSON 11.7

BARBIE BUNGEE – PART 1



LESSON INTRODUCTION

MA.8.DP.1.1 Given a set of real-world, bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

LEARNING TARGETS

- I can collect data for an experiment.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical, and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways, state whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.

ESSENTIAL QUESTIONS

- What kind of data can be modeled with a scatter plot?
- Why do I make predictions or hypothesize before completing an experiment?
- How does changing one test variable affect the data?

LESSON NARRATIVE

Get ready – Barbie is going bungee jumping. This is the first of two lessons where students experiment in order to collect their own data. Students will measure the length of their doll, hypothesize about jumping distances, and then complete the activity, sending Barbie bungee jumping using different amounts of rubber bands to control how far she falls.

COMPONENT SUMMARY

11.7.1 WARM-UP (~5 MIN.)

Students look at how fear can be a part of a growth mindset

11.7.2 ACTIVITY (35-40 MIN.)

Hypothesize and then complete experiment collecting data on the length of Barbie's fall

11.7.3 WRAP-UP (~5 MIN.)

Reflection on student effort and cooperation in the lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- 13 rubber bands per group
- 1 Barbie doll per group
- Rulers with centimeters
- Visible marking of the jump line in the classroom

ADDITIONAL PREPARATION

- 11.7.1 Warm-Up contains a video to accompany the question prompts

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have difficulty identifying the independent and dependent variables for the activity.
- Students may misunderstand how Barbie is to jump from the labeled jumping line.

THINGS TO CONSIDER

- Students take their average for each set of rubber bands to streamline their data.
- Students should all be measuring in centimeters.

LITERACY AND LANGUAGE ROUTINES

Three Reads

As students prepare to send Barbie bungee jumping, consider using the technique of Three Reads to assist in student understanding of what they will be doing and how they will be doing it. By promoting discussion among groups and having a thorough reading of the directions, students may feel more confident in completing the experiment.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

Today's lesson begins and ends with growth mindset reflections and questions. Consider extending the idea of the warm-up and previewing the wrap-up by discussing how a growth mindset will assist them during their experiment, both individually and as they work with their groups.

DIFFERENTIATE INSTRUCTION

This lesson provides students with the opportunity to learn and experiment through movement. Students are able to do the experiment, to be a part of it, and therefore to really see and understand the data they are recording. Participating in the experiment provides visual, kinesthetic, and auditory entry points as students work collaboratively to collect the data.

11.7.1 WARM-UP: GROWTH MINDSET

TEACHER MOVES

Consider using this warm-up as a discussion starter that can also tie in some of the phrases from the wrap-up. How can a growth mindset be used individually, and how can it be used when working in groups?



11.7.1 Warm-Up: Growth Mindset

Fear is a natural emotion but can be difficult to overcome when trying new things. The fear of trying something new sometimes fills us with doubt about our own abilities. Negative thoughts make giving up easy. However, with effort and perseverance, goals are achievable.

1. How does the statement display a growth mindset?

Answers will vary.

Sample response: Growth mindset means believing that you can grow and get better at something rather than giving up. This statement displays growth mindset by expressing you can push through and achieve your goals with hard work.

2. Describe a time when you were fearful before trying or starting something new but did it anyway.

Answers will vary.

3. How did it feel to overcome your fear?

Answers will vary.

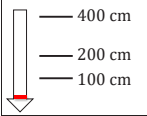
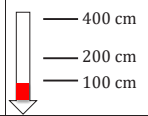
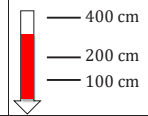


11.7.2 Activity: Barbie Bungee – Part 1

Barbie and her friends are going bungee jumping, but safety is key! Conduct trials to help your team determine the number of rubber bands needed for Barbie to have a safe jump and thrilling bungee experience. Your teacher will explain the process.

Pre-Experiment

- 1. Measure the height of your Barbie in centimeters (cm).
Answers will vary.
- 2. Represent three different risk levels by coloring in each arrow from the top of the jump to the lowest point Barbie’s head is from the ground.
Answers will vary.

Risk Level: Blue	Risk Level: Green	Risk Level: Red
Jump is safe but not thrilling.	Jump is thrilling and safe.	Jump is thrilling but a serious safety issue.
		

- 3. Predict the maximum number of rubber bands needed for Barbie to achieve a jump in the green risk level.
Answers will vary.

11.7.2 ACTIVITY: BARBIE BUNGEE – PART 1

DIFFERENTIATE INSTRUCTION

Consider modeling at least one jump, explaining orally, so that students may get a better grasp on how they are completing the activity.

11.7.2 ACTIVITY: BARBIE BUNGEE – PART 1 (CONTINUED)

TEACHER MOVES

Three Reads

After reviewing as a class, consider implementing the language routine, Three Reads, before beginning the experiment. Students can read through the directions multiple times and then discuss before coming to a consensus about what they will be doing and how they will be doing it.

DIFFERENTIATE INSTRUCTION

Consider having multiple marked “jump lines” clearly visible around the room so that groups may be able to work at the same time.

ENRICHMENT

To deepen student understanding while completing the activity, consider offering guidance:

- Why would we want to measure Barbie’s length?
- How would that help us predict the appropriate lengths? How do you think that will affect the bungee jump length?
- Are there any other variables that might affect the jump?
- Will each group get the exact same data? How do you know?

Data Collection

For the most accurate data, ensure that:

- Barbie is jumping from the specified jump line.
 - A team member is ready to mark the lowest point from each jump.
 - All measurements are in centimeters.
 - All data from jump malfunctions are disregarded and the trial is repeated.
4. Perform three trials for each number of rubber bands with your team.
- a. Record your results from each trial in the table.

Answers will vary.

Number of Rubber Bands Used	Jump Distance (cm)			AVERAGE OF TRIALS (cm)
	TRIAL 1 (cm)	TRIAL 2 (cm)	TRIAL 3 (cm)	
3	50	53	47	50
5	Varies	Varies	Varies	Varies
8	Varies	Varies	Varies	Varies
10	Varies	Varies	Varies	Varies
11	Varies	Varies	Varies	Varies
13	Varies	Varies	Varies	Varies

- b. Calculate the average of the trials for each number of rubber bands and record it in the table.
- Answers will vary.*
5. Consider the quantities being compared in this experiment.
- a. Identify the two variables represented in the table.
- (I) Number of rubber bands used*
(D) Jump distance (cm)
- b. Label the variables identified in Part A as independent (I) and dependent (D).
- c. Explain how you determined the appropriate labels in Part B.

Answers will vary.

Sample response: The jump distance of the barbie depends on the number of rubber bands used.



11.7.3 Wrap-Up: Growth Mindset

1. Reflect on your mindset today by choosing the frequency that you demonstrated the behavior described in each statement.

Answers will vary.

Today I listened to my teammates and I was open to discussing how to complete the work.	☐ Never ☐ Sometimes ☐ Usually ☐ Always
I thought of today's lesson as a learning opportunity instead of rushing to complete the work.	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Today I asked for clarification when I was unsure about something.	☐ Never ☐ Sometimes ☐ Usually ☐ Always
I put as much effort as I possibly could into today's work.	☐ Never ☐ Sometimes ☐ Usually ☐ Always

2. Based on your responses, what will you do to improve or maintain a growth mindset tomorrow?

11.7.3 WRAP-UP: GROWTH MINDSET

DIFFERENTIATE INSTRUCTION

Consider using student responses and observation from today's lesson to inform of which groups may need more guidance or observation when completing the activity tomorrow.